

Newton's Law of Cooling/Warming In equation (3) of Section 1.3 we saw that the mathematical formulation of Newton's empirical law of cooling/warming of an object is given by the linear first-order differential equation

$$\frac{dT}{dt} = k(T - T_m), \quad (2)$$

where k is a constant of proportionality, $T(t)$ is the temperature of the object for $t > 0$, and T_m is the ambient temperature—that is, the temperature of the medium around the object. In Example 4 we assume that T_m is constant.

EXAMPLE 4 Cooling of a Cake

When a cake is removed from an oven, its temperature is measured at 300°F . Three minutes later its temperature is 200°F . How long will it take for the cake to cool off to a room temperature of 70°F ?

14. A thermometer is taken from an inside room to the outside, where the air temperature is 5°F . After 1 minute the thermometer reads 55°F , and after 5 minutes it reads 30°F . What is the initial temperature of the inside room?
15. A small metal bar, whose initial temperature was 20°C , is dropped into a large container of boiling water. How long will it take the bar to reach 90°C if it is known that its temperature increases 2° in 1 second? How long will it take the bar to reach 98°C ?

Expl 1: When a cake is removed from an oven. Its temperature is measured at 300°F . Three minutes later its temperature is 200°F . How long will it take for the cake to cool off to a room temperature of 70°F .

Solⁿ:

$$T(0) = 300^\circ\text{F}$$

$$T(3) = 200^\circ\text{F}$$

$$T(t) = 70^\circ\text{F}$$

$$\therefore \frac{dT}{dt} = K(T - 70)$$

$$\Rightarrow \int \frac{dT}{T - 70} = K \int dt$$

$$\Rightarrow \ln |T - 70| = Kt + C$$

$$\Rightarrow T = 70 + Ce^{Kt}$$

$$T(0) = 300$$

$$\Rightarrow 70 + C = 300$$

$$\Rightarrow C = 230$$

$$T(3) = 70 + 230e^{3K} = 200$$

$$\Rightarrow 230e^{3K} = 130$$

$$\Rightarrow e^{3K} = \frac{13}{23}$$

$$\Rightarrow 3K = \ln \frac{13}{23}$$

t	T
0	300
3	200.02
10	104.4
20	75.15
30	71.77

$$\begin{aligned}\Rightarrow e^{2k} &= \frac{13}{23} \\ \Rightarrow 2k &= \ln \frac{13}{23} \\ \Rightarrow k &= \frac{1}{2} \ln \frac{13}{23} \\ &= -0.19\end{aligned}$$

$$\begin{array}{c|c} 30 & 71.77 \end{array}$$

14 $T_{air} = 5^\circ F$

$$T(1) = 55^\circ F$$

$$T(5) = 30^\circ F.$$

Here, $\frac{dT}{dt} = k(T - T_{air})$

$$\Rightarrow \int \frac{dT}{T-5} = \int k dt$$

$$\Rightarrow T = 5 + ce^{kt}$$

For, $T(0) = 5 + c$. ——— (i)

For, $T(1) = 5 + ce^k$

$$\Rightarrow 50 = ce^k$$

$$\Rightarrow c = \frac{50}{e^k}$$
 ——— (ii)

For, $T(5) = 5 + ce^{5k}$

$$\text{For, } T(\bar{y}) = \bar{y} + ce^{5k}$$

$$\Rightarrow 2\bar{y} = ce^{5k}$$

$$\Rightarrow C = \frac{2\bar{y}}{e^{5k}} \longrightarrow \textcircled{\text{iii}}$$

From $\textcircled{\text{i}}$ and $\textcircled{\text{ii}}$,

$$\frac{\bar{y}_0}{e^k} = \frac{2\bar{y}}{e^{5k}}$$

$$\Rightarrow e^{4k} = \frac{1}{2}$$

$$\Rightarrow 4k = \ln\left(\frac{1}{2}\right)$$

$$\Rightarrow k = \frac{1}{4} \ln\left(\frac{1}{2}\right)$$

$$= -0.173$$

$$\therefore C = 59.40$$

$$\begin{aligned} T(0) &= 5 + c \\ &\Rightarrow 64.46 \quad (\text{Answer}) \end{aligned}$$

17) Given, $T(0) = 20^\circ\text{C}$
 $T(1) = 22^\circ\text{C}$

$$T_{bp} = 100^\circ\text{C}$$

$$\frac{dT}{dt} = k(T - 100)$$

$$\Rightarrow \int \frac{dT}{T - 100} = \int k dt$$

$$\Rightarrow T = 100 + ce^{kt}$$

$$\text{Now, } T(0) = 100 + ce^{k0}$$

$$\Rightarrow 20 = 100 + ce^{k0}$$

$$\Rightarrow C = -80$$

$$\text{Now, } T(t) = 100 + Ce^{kt}$$

$$\Rightarrow 22 = 100 - 80e^k$$

$$\Rightarrow k = -0.0253$$

$$\therefore T(20^\circ\text{C}) = 100 + Ce^{kt_1}$$

$$\Rightarrow 20 = 100 + (-80)e^{(-0.0253)t_1}$$

$$\Rightarrow t_1 = 82.2 \text{ sec (Ans.)}$$

$$\therefore T(98) = 100 + Ce^{kt_2}$$

$$\therefore T(98) = 100 + Ce^{kt}$$

$$\Rightarrow 98 = 100 + (-80)e^{(0.0253)t_2}$$

$$\Rightarrow t_2 = 145.805512 \text{ sec (Ans.)}$$